

Wireless charging

Advantages

- Low maintenance as there are no moving parts
- Non-wired connection makes it safer as it eliminates the risk of short circuit from any damaged wire or connectors
- Eliminates the need to carry charging accessories, such as chargers and charging cable
- Improves water and dust resistance of a device as there are no ports

Drawbacks

- The rate of power transfer is much lower than the wired method of charging
- Efficiency of wireless charging is lower than regular wired charging
- Charging distance is limited
- The charging coils need to align perfectly to charge efficiently in case of inductive charging

solutions enable higher transmitter and receiver frequencies of 6.78MHz or 13.56MHz, which are ideal for resonant charging.

The main advantages of the adoption of GaN FET are:

- Higher efficiency
- Lower cost
- Reduction in weight and size of the system
- Lower surface heat loss

Long distance charging

RF charging is hailed as the future of wireless energy harvesting for low-power IoT devices. The future scope of RF wireless charging was discussed at CES 2022. This charging technology allows charging at long distances from the transmitter.

At CES 2022, Motorola demonstrated over-the-air wireless charging of four phones simultaneously from a single transmitter for distances as far as 122cm (4 feet). Xiaomi and Oppo demonstrated their RF charging prototypes. Samsung also revealed an RF charging enabled remote, which was also featured in Best of CES 2020.

The RF frequency charging is capable of harnessing different frequencies from ambient energy. The technology is expected to mature in the coming years. RF charging is considered most suitable for smart wearable devices, IoT sensors, and applications in medical, military, sports, and many other industries.

The future

Wireless charging technology will give engineers the freedom to make devices smaller and lighter by eliminating the energy storage devices, such as bulky batteries, in them. The improvement in processing capabilities of our handheld devices makes wireless charging a convenient or rather an essential technology.

Wireless charging is not limited to consumer electronics, it also has scope in automotive industry. Recently, in March 2022, Volvo announced that its XC-40 model will be able to receiving power greater than 40kW, which is much more than the existing AC charging and almost comparable to the DC fast charging.

RF charging is the key to making batteryless devices. Thus, wireless charging holds a lot of scope in commercial, medical, smart wearable, and military applications. Further development in this technology will help us get rid of chargers, cables, and power banks. We may soon witness the adoption of wireless charging solutions in the public space. RF charging will enable sensors placed at remotest of the location to work perfectly without human intervention. **EFY**

The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world.



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